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ORIGINAL RESEARCH

Mitigating risk in medical AI: balancing X-ray datasets for reliable detection

Mohd Aquib Ansari[1] · **Shahnawaz Ahmad**[1] · **Arvind Mewada**[1]

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Abstract
A significant challenge in medical healthcare is the limited availability of labelled imaging data, coupled with class imbalance issues, which adversely affect the performance of learning algorithms reliant on large, high-quality datasets. In many cases, datasets contain unequal disease and non-disease cases, leading to biased models that struggle with generalisation. Therefore, this research explores the Deep Convolutional Generative Adversarial Networks (DCGANs) to generate synthetic chest X-ray images, dealing with the issue of class imbalance. The system is further trained using the ResNet50 model for the augmented chest X-ray image dataset labelled either pneumonia or normal. Evaluation using DCGANs with ResNet50 demonstrated a success rate of 93.4% in producing photorealistic images, validating their potential to augment existing datasets. The synthetic images facilitate improved model training, enhancing diagnostic accuracy in medical facilities by providing enriched medical imaging datasets. This study highlights how advanced generative methods can address healthcare data constraints, enabling precise and efficient diagnostic systems.

Keywords Deep convolutional GAN (DCGAN) · Medical imaging · Data augmentation · Image synthesis · Medical data scarcity

1 Introduction

Advanced medical imaging technologies play a pivotal role in modern healthcare, enabling accurate disease diagnosis, treatment planning, and monitoring of illness progression. Medical professionals rely on imaging modalities such as X-rays and Magnetic Resonance Imaging (MRI) to make informed clinical decisions. However, the growing demand for automated medical image interpretation has created a need for large, high-quality datasets. Machine learning (ML) and deep learning (DL) models (Alshanketi et al. 2024; Goodfellow et al. 2014) depend on these datasets for training and validation, significantly enhancing disease classification, segmentation, and anomaly detection. On the other hand, DL has demonstrated substantial progress in medical imaging and its success hinges on the availability of diverse and high-resolution datasets. The painstaking steps behind the acquisition and the annotation processes are the two primary areas of concern regarding possessing enough medical image data, alongside ethical restraints and privacy issues. This becomes even more complex given that medical datasets usually have an imbalance when underrepresenting certain conditions, such as normal cases, while simultaneously overrepresenting pneumonia, causing predictions to be biased and the model's generalisation to be weak.

Modern diagnostics still heavily rely on medical imaging. Yet, most solutions that use artificial intelligence to address healthcare needs face some difficulties due to the lack of class balance issues (Sharma and Gosain 2025) and insufficient data. The labelled datasets for medical imaging tend to be scarce because of the data sharing restrictions present, the high costs of manual annotation and the infrequent occurrence of some pathological conditions. Data expansion for augmentation methods has been done using flipping, rotating, and altering the contrast, but such methods do not provide real anatomical or disease pattern

✉ Arvind Mewada
arvind.mewada@bennett.edu.in

Mohd Aquib Ansari
mohd.aquib@bennett.edu.in

Shahnawaz Ahmad
shahnawaz98976@gmail.com

[1] School of Computer Science, Engineering and Technology, Bennett University, Greater, Noida, India

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variation. These methods also fail at attempting to solve the class imbalance problem, which has the potential to bias the model's predictions. However, the recent advancements in Generative Adversarial Networks (GANs) (Goodfellow et al. 2014) greatly aid this issue. They can create high-quality synthetic images imitating real medical scans while balancing the dataset.

Despite the increased use of advanced learning techniques to analyse medical imaging, several research gaps remain. One of the most significant problems is the scarcity of medical datasets, with most of the prior research relying on undersized, biased datasets that do not possess the necessary multiformity for effective model training. Moreover, although traditional data augmentation methods help improve the dataset's size, they seldom provide realistic changes that would appropriately model actual medical scenarios. Another area that has not been adequately studied is using GANs in medical imaging, emphasising pneumonia detection and chest X-ray imaging. While the use of GANs for generating synthetic images has been explored, their application in these areas has not yet been fully grasped. In addition, there is a gap in the literature concerning the systematic evaluation of the role of synthetic data in DL model performance, with scant attention given to the analysis of images generated by GANs and real-life images. Tackling the class imbalance issue is still an open question since most DL algorithms, which are trained with imbalanced data, have a very limited scope of generalisation and are not very useful in practical medicine (Darapaneni et al. 2022; Heusel et al. 2017).

This study aims to enhance the accuracy of pneumonia detection via DCGANs by improving the quality and availability of medical imaging datasets, such as chest X-rays. This method creates synthetic X-ray pictures of the chest while solving class imbalance issues by generating samples for underrepresented classes. The proposed augmentation method ensures adequate samples from all classes in the training dataset, thus improving the DL models' accuracy in classifying medical images. This research addresses the issues of class imbalance and the paucity of available datasets through the integration of DCGANs into the data augmentation pipeline, resulting in improved accuracy of diagnosis and better generalisation of the model.

We make several important contributions to this study. First, we develop and implement a DCGAN architecture for generating high-resolution synthetic chest X-ray images for an underrepresented X-ray image classification. Using these synthetic images improves data diversity and ensures adequate representation of all classes, thus enhancing the robustness of DL models. In addition, we analyse the classification performance of DCGAN-generated images for pneumonia using the ResNet50 model, aiming to understand their value added in classification accuracy. A growing body of research argues for using GANs in the augmentation of medical images compared to traditional data augmentation methods that do not take advantage of generative models. We show through experiments how the addition of synthetic images contributes to better model generalisation and higher accuracy of pneumonia detection, thereby improving the quality of diagnostic results.

The rest of the document is structured in the following manner. A literature review is covered in section II. The proposed methodology is presented in section III with specific details on the DCGAN architecture, dataset, the ResNet50 classification model, and training strategies. Results of the experiments are provided in section IV, where the quality assessment of the synthetic images and their effect on the classification performance of pneumonia are shown. Section V is devoted to concluding remarks and outlines the study's contributions, its implications in the analysis of medical images, and advances which may be made in the future augmentation of the data using GANs.

2 Related work

The pneumonia detection process significantly benefits from the techniques of DL, which has greatly improved the field of medical image analysis. Researchers have tried different ways to improve the accuracy of health data analysis and tackle the scarcity of health data. Some key methods are generating synthetic data, transferring learning, and using computational models based on convolutional neural networks (CNN).

One of the latest research works conducted by Poola et al. (2024) presented a new approach, CE-GAN (Classifier-Embedded Generative Adversarial Network), which used the feature extraction process of a medical image classification pipeline to incorporate synthetic medical images. The model certainly performed better due to the availability of artificial data. Mendes et al. (2023) conducted a comparative experiment to confirm the received results and analyse the effectiveness of using GAN-generated datasets to improve the analysis of medical images. Our work extends these results using Deep Convolutional Generative Adversarial Networks (DCGAN) to produce synthetic chest X-ray images. This allows for tackling class imbalance problems while enabling the model to detect pneumonia more proficiently.

Avola et al. 2022 compared several pre-trained pneumonia detection models, showing that transfer learning considerably increases classification accuracy and decreases computation costs. Similarly, Gu and Lee 2024 utilised features from ImageNet and showed that using pre-trained

models is beneficial in medical imaging. This research uses GAN images produced alongside transfer learning to improve the quality of the training dataset and, therefore, the adaptability and robustness of the model.

Besides transfer learning, lightweight CNN architectures have performed exceptionally in pneumonia detection. Dabre et al. 2024 implemented a low-cost CNN model, which accomplished 94.6% accuracy while not overly complex. Meanwhile, Garstka and Strzelecki 2020 applied data augmentation techniques to CNN models, enabling the classification of X-ray images into three categories: healthy, bacterial pneumonia, and viral pneumonia. Expanding on these findings, our research incorporates DCGAN-generated synthetic images into CNN-based approaches. This integration has led to a notable 91.2% accuracy improvement over traditional CNN methods.

Further advancements in pneumonia detection have been made through ensemble CNN models and multi-channel feature fusion techniques. Bhatt and Shah 2023 introduced an ensemble CNN model with multiple kernel sizes, balancing high recall with interpretability. Similarly, Vieira et al. 2021 explored multi-channel feature fusion, incorporating Local Binary Patterns (LBP) and Contrast-Limited Adaptive Histogram Equalization (CLAHE) to enhance feature extraction. While our study does not explicitly apply feature fusion techniques, using GAN-generated synthetic images introduces greater variability in training data, improving overall model robustness.

While previous research has predominantly focused on CNN-based feature extraction, transfer learning, and ensemble models, our study uniquely integrates DCGAN-generated synthetic data to strengthen training datasets, mitigate class imbalance, and enhance pneumonia detection accuracy. Our approach advances DL applications in medical imaging by leveraging GAN-based augmentation. The comparative analysis of existing research findings is presented in Table 1.

3 Methodology

The block diagram, as shown in Fig. 1, illustrates the workflow of a DCGAN+ResNet pipeline for pneumonia classification using chest X-ray images. The process begins with data collection, where a chest X-ray dataset is gathered, often suffering from class imbalance, which can hinder classification performance. To tackle this issue, we use DCGAN (Fang et al. 2018) for data augmentation and to create synthetic X-ray images to balance the dataset and improve generalisation performance. Originally captured images alongside synthetic ones constitute the new dataset, which is then passed through ResNet, a deep convolutional neural network with residual learning, for feature extraction and classification. ResNet processes the images seamlessly and classifies them into normal and pneumonia cases. During testing, the trained chest X-ray ResNet model predicts classification outcomes for unseen images: pneumonia or normal. After the final evaluation phase, which establishes model performance using accuracy, precision, recall, and F1 score as key measures, results show that the model, also known as ResNet, captured the bottom features very well when trained on the augmented data (DCGAN+ResNet). This model outclassed conventional CNN models because it was able to balance severe class imbalance, improve feature extraction, and enhance classification performance. The model successfully addresses the dataset imbalance, making

Table 1 Comparative analysis of existing approaches

Reference	Method Used	Dataset	Key Findings	Limitations
Poola et al. (2024)	Classifier-Embedded GAN (CE-GAN)	Medical image dataset	Synthetic images improved classification accuracy	Potential overfitting to synthetic data
Mendes et al. (2023)	GANs for dataset enhancement	Medical imaging datasets	GANs improved dataset quality and model performance	Requires careful evaluation of synthetic image realism
Avola et al. (2022)	Transfer learning with pre-trained models	Pneumonia detection dataset	Transfer learning enhances classification accuracy while reducing the computational cost.	Requires large-scale pre-trained models
Gu and Lee (2024)	ImageNet-based feature representations	Medical imaging dataset	Pre-trained models enhance performance in medical imaging tasks	It may not capture domain-specific medical features
Dabre et al. (2024)	Lightweight CNN model	Medical X-ray dataset	Low computational complexity with high accuracy	May not generalise well to complex cases
Garstka and Strzelecki (2020)	CNN with data augmentation	X-ray dataset	Data augmentation improves CNN classification	Limited augmentation techniques compared to GANs
Bhatt and Shah (2023)	Ensemble CNN with multi-kernel sizes	Medical image dataset	Achieves high recall while maintaining interpretability	Computationally intensive
Vieira et al. (2021)	Multi-channel feature fusion (LBP, CLAHE)	Medical imaging dataset	Feature fusion enhances model robustness	Complexity increases with additional feature channels
Our Study	DCGAN-generated synthetic images	Chest X-ray dataset	Reduces class imbalance and enhances model performance	Mode collapse risk requires careful tuning

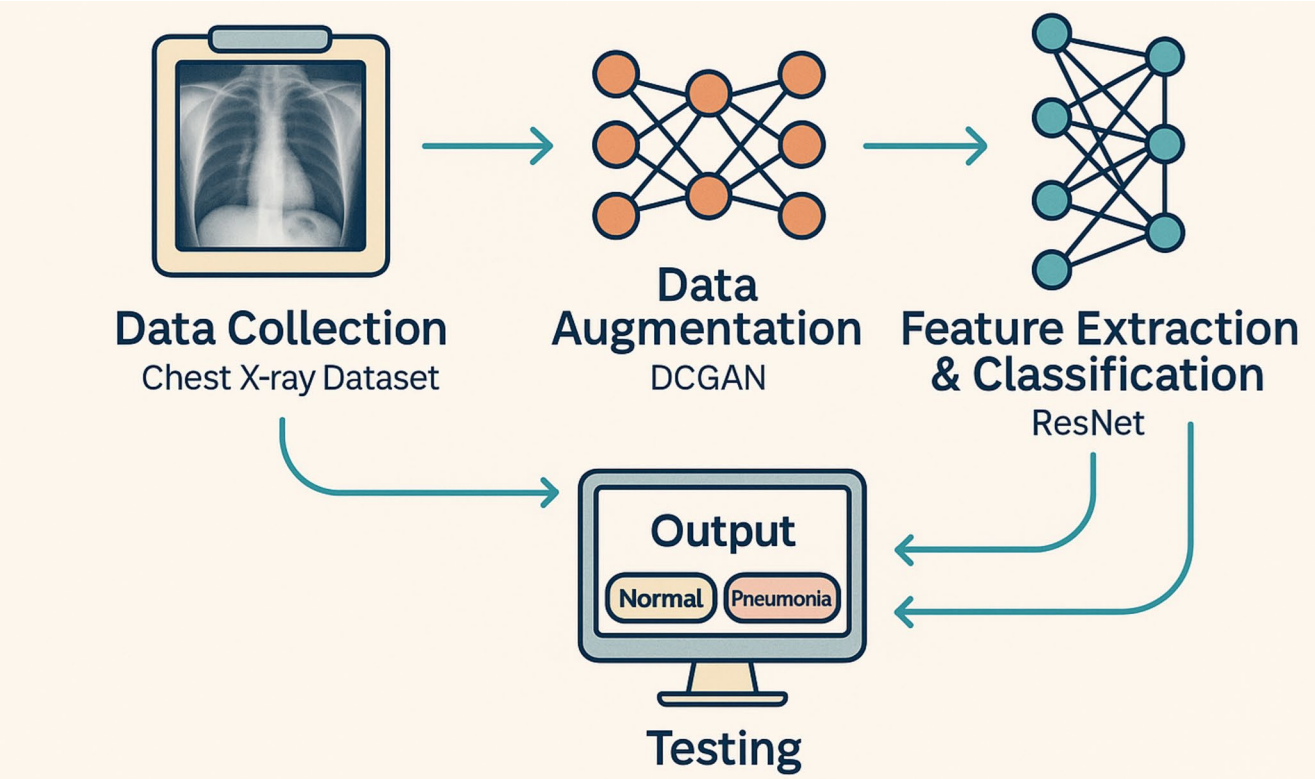


Fig. 1 Comprehensive Pipeline of Data Augmentation and Deep Learning-Based Pneumonia Diagnosis

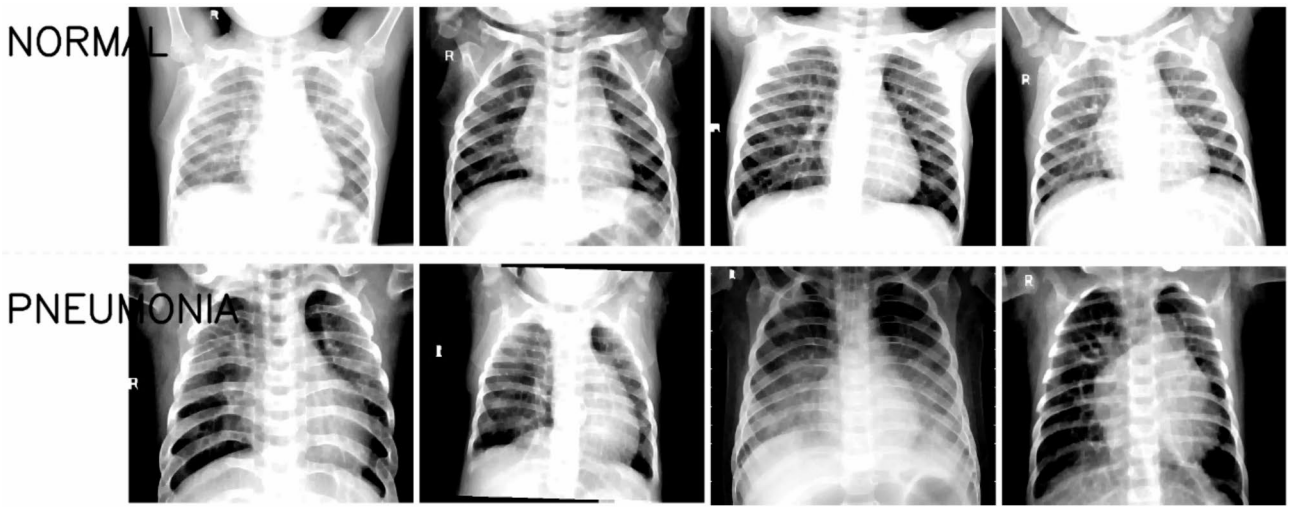


Fig. 2 Sample of chest Xray dataset

the pneumonia detection system more accurate and dependable, making it ideal for practical use in the medical field.

3.1 Data collection

Chest-xray pneumonia dataset (Mooney 2023) includes 5863 grayscale chest X-ray images taken from an anterior-posterior view. The X-rays in the dataset are stratified into two classes labelled 'NORMAL' and 'PNEUMONIA.' It is further stratified into a training portion and a testing portion. The training set has a total of 5216 images, 1341 of which are NORMAL and 3875 of which are PNEUMONIA images. The remaining 624 images in the test set, 134 being NORMAL and 387 being PNEUMONIA. There is a class imbalance in this dataset as there are significantly more cases of pneumonia than normal cases. Some instances of the same dataset are represented in Fig. 2, where the first row shows samples of the normal class, and the second row shows the

pneumonia class. The NORMAL X-rays show clear lung fields with no significant abnormalities. On the other hand, the PNEUMONIA X-rays present increased denseness and structural modifications due to infection. Radiological diagnosis of pneumonia benefits from this classification system in detecting the disease.

A DCGAN is used here to create synthetic images for the underrepresented 'NORMAL' class, which also helps mitigate the imbalance. The details related to DCGAN are shown in the next subsection. Instead of using traditional data augmentation that transforms existing images, DCGAN moves away from such methods. It passes the complicated data distribution of real chest X-ray photos and creates new samples. In total, 2534 synthetic NORMAL images were produced and added to the training dataset, which helps achieve a better balance in class distribution. Now, this augmentation aids DL models in becoming more robust because these models will be trained with diverse data, which is important to make the model generalise better and perform classification tasks effectively. The final augmentation set created contains 7750 training images (3875 NORMAL and 3875 PNEUMONIA) while keeping the class split of 90%-10% train-test.

Including images generated by a DCGAN was fundamental to enhancing pneumonia detection by adding more synthesised images to the existing dataset. These images underwent assessment through qualitative and quantitative analyses, and their accuracy and usefulness in augmenting real data were validated. The refined dataset, which is inclusive, well-distributed, and diverse, is beneficial in developing stronger DL models. This transforms the accuracy and reliability of the model in real clinical settings, making it more advanced in diagnosis and model performance.

3.2 DCGAN model design and implementation

The heart of the method is centred on creating and applying the DCGAN. DCGANs belong to the group of GANs that implement deeper convolutional-based networks in both the generating and discriminating modules to guarantee the generation of high-quality media. A visual depiction in Fig. 3 demonstrates how the DCGAN components depend on one another. The Generator commences operations using Z+C and then transforms this vector in convolutional and upsampling layers to produce a $64 \times 64 \times 3$ image output. The Discriminator evaluates either real or generated images for analysis through its convolutional layers. A final fully connected layer within the Discriminator enables it to determine real or synthetic image classes. The Generator creates realistic images, but the Discriminator becomes skilled at differentiating real images from generator-generated images. The detailed explanation concerning DCGAN consists of the following points.

3.2.1 Generator architecture

The generator oversees the production of synthetic media from random noise vectors. Its architecture comprises several transposing convolutional layers designed to upsample latent space vectors (usually sampled from a normal distribution) into a high-dimensional image. The generator includes the following key components.

- **Input Layers**: The generator's input is a random vector (z) of size 100, sampled from a normal distribution.

$$z \sim \mathcal{N}(0, I) \tag{1}$$

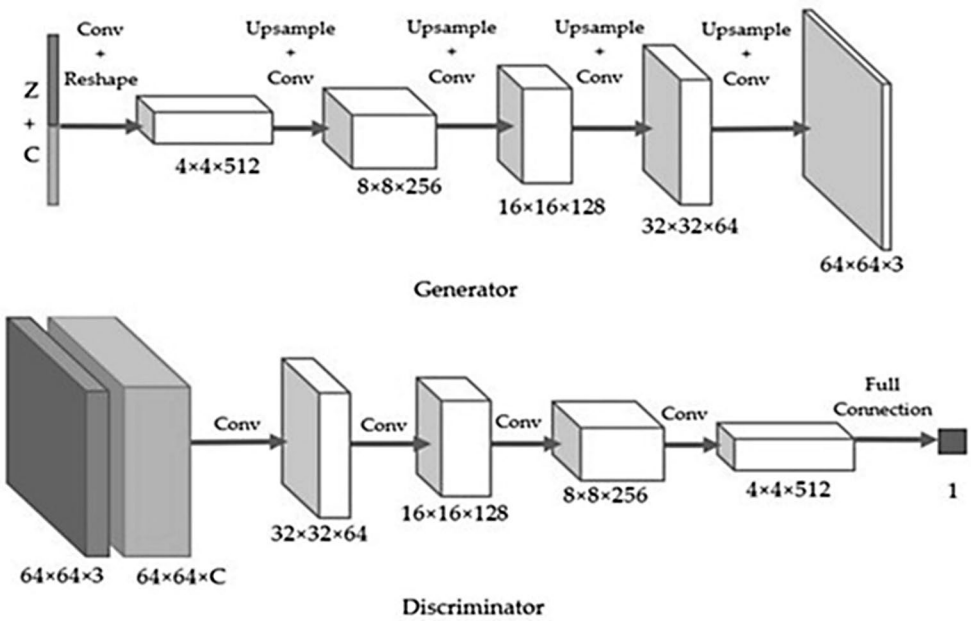


Fig. 3 DCGAN layered architecture pipeline

where z is a 100-dimensional latent space vector.

- **Dense Layer**: The first layer of the generator is a fully connected layer that reshapes the input vector into a $4 \times 4 \times 512$ feature map.

$$h_0 = W_0 z + b_0 \tag{2}$$

Where the weight matrix is represented by W_0, z is biased, and h_0 represents a feature map of size $(4 \times 4 \times 512)$.

- **Transposed Convolutional Layers**: These layers upsample the feature map to increase the spatial resolution of the output. The number of filters progressively decreases as spatial resolution increases.

$$h_i = f\left(W_i * h_{i-1} + b_i\right) \tag{3}$$

Where W_i, $*$, b_i, and f represents convolutional kernel weights, transposed convolution operation, bias, and activation function, respectively.

- **Batch Normalisation and Leaky ReLU**: Batch normalisation is applied after each convolutional layer to optimise training. The Leaky ReLU activation function prevents dying neurons and ensures smooth gradient flow.

$$\widehat{x_i} = \frac{x_i - \mu}{\sigma + \varepsilon} \tag{4}$$

Where μ and σ are mean and variance, ε and $\widehat{x_i}$ are constant and normalised activation, respectively.

- **Output Layer**: The final layer applies the tanh activation function, scaling pixel values between [-1,1] to ensure proper normalisation.

$$\widehat{x} = \tanh\left(W_n * h_{n-1} + b_n\right) \tag{5}$$

3.2.2 Discriminator architecture

The discriminator is a binary segregator that differentiates between real and generated media. It is the result of a few convolutional layers that progressively reduce the feature sizing of the input media. The discriminator consists of the following components.

- **Convolutional Layers**: The early layers consist of standard 2D convolutional layers with strides larger than one, reducing spatial resolution while increasing the number of filters. The model maintains the capability to detect features of both complex and simple nature in the input visual data.
- **Leaky ReLU Activation** deals with vanishing gradients by providing minimal negative slopes when negative inputs exist to maintain stable training conditions.

$$f(x) = \begin{cases} x & if\ x \geq 0 \\ ax & if\ x < 0 \end{cases} \tag{6}$$

Where a is a small negative slope coefficient.

- **Batch Normalisation** enables stable training and better convergence when applied to post-convolutional layers.
- The combination of **fully connected layer** technology produces one output value that indicates the likelihood of an image being authentic or artificial.
- **Sigmoid Activation**: The final output passes through the sigmoid activation function, mapping values to the range [0,1], where 1 indicates a real image, and 0 represents a generated (fake) image.

$$D(x) = \sigma\ (y) = \frac{1}{1 + e^{-y}} \tag{7}$$

Where $\sigma\ (y)$ is sigmoid activation, and y represents a fully connected layer.

3.2.3 Adversarial loss function

The training of the DCGAN takes place simultaneously over two objectives, which are defined as follows:

- **Generator Loss**: The generator's objective is to minimise the cross-entropy loss engaged in by the discriminator, making D(G(z)) trick the discriminator into nominating images as real captures. G(z) captures the image produced by the generator, and D (G(z)) denotes the probability that the discriminator classifies a generated image as a real capture.

$$L_G = -E_{z\sim\, p_z(z)}\left[\log D\left(G\left(z\right)\right)\right] \tag{8}$$

- **Discriminator Loss**: The discriminator aims to misclassify nominal images D(x) as real images to maintain its prediction for captured images G(z). D(G(z)) denotes the discriminator's prediction for the generated images. The loss function is the summation of the cross-entropy loss over real and nominal images.

$$\begin{aligned} L_D = &-E_{x\sim\, p_{data}(x)}\left[\log D\left(x\right)\right] \\ &- E_{z\sim\, p_z(z)}\left[\log\left(1 - D\left(G\left(z\right)\right)\right)\right] \end{aligned} \tag{9}$$

3.2.4 Similarity evaluation

The evaluation method for assessing generated and real images uses Frechet Inception Distance (FID). A reduced FID measurement indicates superior quality in the generated imagery.

$$FID = ||\mu_r - \mu_g||^2 + Tr\left(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2}\right) \quad (10)$$

3.2.5 X-Ray image classification using ResNet50

ResNet-50 (Residual Network with 50 layers) is a deep convolutional neural network implemented to solve the vanishing gradient problem by introducing residual connections. These skip connections allow the efficient training of deep networks as they enable the flow of gradients using identity shortcuts, which mitigates the loss of information due to backpropagation. The ResNet-50 architecture is illustrated above; the black edges are the connections of layers in order, while the red edges are skip connections that allow for jumping over multiple layers and aid in reusing already extracted features. The model first has a convolutional layer (Conv1) followed by a max-pooling layer (MaxPool). Multiple residual blocks then process the data. Convolutional layers of each residual block have identity shortcuts that aid in some feature representations, such as important ones being captured. The ability to skip connections helps to learn since having deeper networks can then be trained without having performance problems because of the networks' depth.

Figure 4 demonstrates that ResNet-50 comprises several residual blocks identified as Res Block 1 to Res Block 12 that deal with hierarchical feature representation. The skip connections marked in red are the identity mappings, where the information within the layer is passed through without any alterations to subsequent layers. The introduced residual connections help deal with the degradation problem associated with traditional deep networks, leading to better convergence and accuracy in the models. To use the model, one must first process the image using the residual blocks and then forward the output to a fully connected layer for classification. RESnet-50 has proven to be the model of choice for many users due to its outperforming ability in the recognition of images and, therefore, has become central in DL development, such as in medical imaging, object detection, and natural classification of images.

4 Results analysis

DL models are trained and tested in a high-performance computing environment to manage the sizable chest X-ray datasets effectively. The system used an NVIDIA RTX 3060 GPU with 12GB VRAM, an Intel Core i7 processor, 16GB of RAM, and a 1 TB SSD, all deployed on Ubuntu 20.04 LTS. Python 3.8 was used to implement the models along with DL frameworks TensorFlow 2.9, PyTorch 1.12, and other important libraries such as OpenCV, NumPy, Pandas, and Matplotlib for data preprocessing and visualisation purposes. Computation time was reduced significantly by enabling GPU acceleration with CUDA 11.6 and cuDNN 8.4 during the training and augmentation. The dataset was

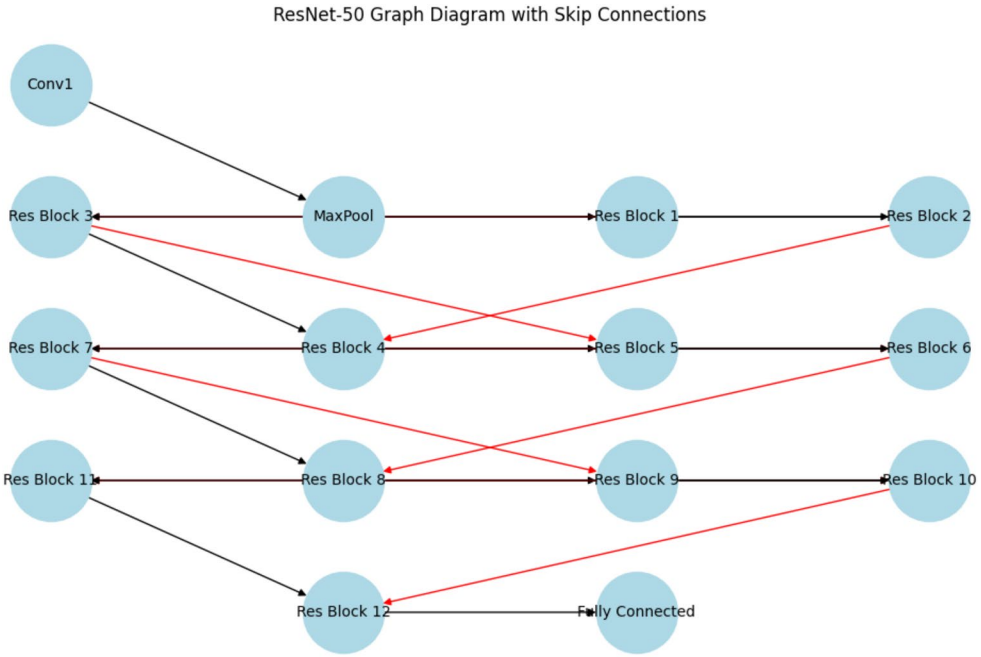


Fig. 4 Resnet50 graph diagram with skip connections

facilitated using TensorFlow Datasets (TFDS) and the PyTorch DataLoader, guaranteeing proper data management. The tests were performed in a controlled environment with automatic mixed precision (AMP) active for faster training speed and robustness to numerical stability problems.

The research used the DCGAN model, with a latent vector with 100 values and the Adam optimizer ($\beta1 = 0.5$ and $\beta2 = 0.999$) at a learning rate of 0.0002 for the generator and 0.0004 for the discriminator. Throughout the operation, the generator applied transposed convolutional layers, batch normalization, leaky ReLU ($\alpha = 0.2$), and tanh activation functions to scale its output values. The discriminator uses a 2D Convolutional Layer along with the Leaky ReLU activation and sigmoid functions for classification tasks. A training process is carried out with a batch size set to 64 while operating for 200 epochs using binary cross-entropy loss to achieve stable convergence and produce high-quality synthetic images.

On the other side, hyperparameters associated with ResNet50 are optimised for the best classification results for pneumonia detection. The model was set up using transfer learning, where pre-trained weights from the ImageNet database were used and later fine-tuned with the model specifically for pneumonia classification. The Stochastic Gradient Descent (SGD) optimiser was used with an initial learning rate of 0.001 and momentum of 0.9 to ensure proper weight updates. The model was trained for 100 epochs with an early stopping criterion, which was assigned based on validation loss to reduce chances of overfitting, applying a 20 epochs patience. The batch size was set to 32, and since the model was a multi-class classifier, categorical cross-entropy loss was used. To strengthen generalisation, data augmentations were applied, such as random rotations, flipping, and changes in brightness. The learning rate was adjusted as required through a ReduceLROnPlateau scheduler with patience set to 25 epochs and a decay factor of 0.1. Furthermore, a dropout of (0.4) was added on the fully connected layers to reduce overfitting and increase the model's robustness.

4.1 Performance impact on pneumonia detection

As shown in Fig. 5, the graph outlines the performance values of ResNet50 with and without DCGAN data augmentation. It can be observed that the addition of the synthetic images greatly improves the model performance (Tiwari et al. 2024; Ansari et al. 2021; Aljawarneh and Romesaa Al-Quraan 2025; Kumar et al. 2023; Yadav et al. 2025) for all important metrics considered—accuracy, precision, recall,

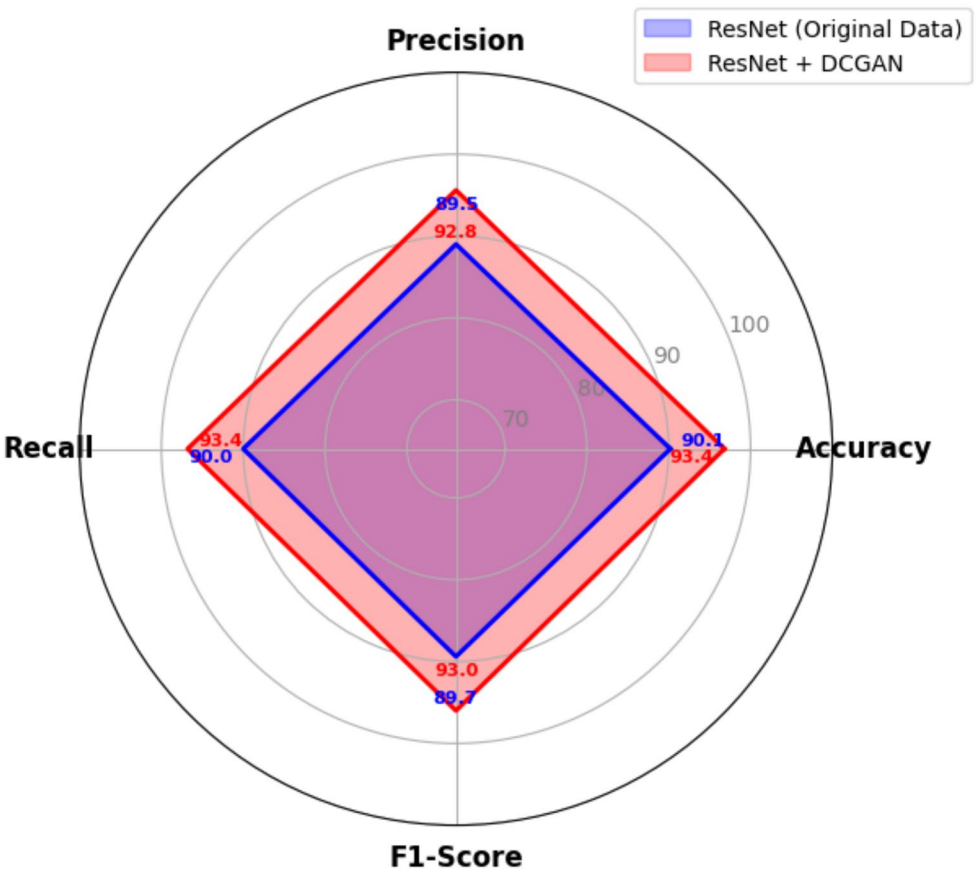


Fig. 5 Comparative Radar Chart for the respective Performance metrics

and F1-score. In particular, the improvement in recall means that the model is better at detecting some pneumonia cases and mischievously failing less. Parallelly, the F1 score, the average of precision and recall, also increases considerably, proving that the model can generalise and perform well. The increase in precision demonstrates that the model can reduce the number of false positive predictions, which renders the predictions trustworthy. In conclusion, amid class imbalance, the DCGAN-generated data aids in attaining a robust classifier, resulting in a dependable and efficient pneumonia detection system. This modification stresses the importance of synthetic data for enhancing model performance and diagnostic accuracy in medical imaging.

The accuracy improvement for ResNet50 with and without DCGAN-based augmentation over 100 epochs is illustrated in the performance graph, as shown in Fig. 6. The results show that ResNet50+DCGAN (in red and dashed line) is always better than baseline ResNet50 (in solid blue) throughout all epochs. Both models show biased accuracy growth at the beginning of the training period. However, the model with DCGAN goes on to achieve a higher final accuracy of around 93.4%, while ResNet50 DCGAN closes in at around 90.1%. This indicates that including synthetic images produced by the DCGAN strengthens the model's generalisation ability and performance in class imbalance tasks. The accuracy improvement further highlights the efficacy of DCGAN augmentation in correcting the underrepresentation of some classes in the training data, leading to better robustness and accuracy of the model.

The augmentation with DCGAN performs with ResNet50, depicted in the training and validation loss curve in Fig. 7, which demonstrates performance over 100 epochs. Both approaches have shown a tempered decline in loss, demonstrating learning achievement over the training cycle. The ResNet50+DCGAN model experiences greater initial loss when set against ResNet50. However, as training proceeds progressively, loss values of both models settle at lower levels, with the DCGAN augmented model exhibiting greater loss. This indicates that even though synthetic data enhances generalisation and improves model performance, it also causes some inconsistency in the learning phenomenon. Regardless, both models have shown remarkable and consistent loss reduction, highlighting the role of DL techniques in classifying pneumonia X-rays. It is conclusively observed that the augmentation using a DCGAN increases the model's ability to withstand perturbations and increases generalisation, which validates the claim.

4.2 Comparison with baseline model

The impact of DCGAN-based augmentation on model performance across different architectures is notable, as presented in Table 2. The baseline CNN (Sequential CNN, containing two convolutional and two max pooling operations along with flattening and softmax layers) accuracy was 82.3%, which increased to 85.7% with the use of DCGAN synthetic data, thus proving the benefits that synthetic data has on the model performance. In the same way, VGG16 and VGG19 increased in accuracy, achieving 89.3% and 90.0%, respectively, with DCGAN augmentation, confirming the effectiveness of DCGAN in feature extraction. The most significant change was seen in ResNet50, which improved accuracy from 90.1 to 93.4%, making it the best model. This pattern holds true for other important metrics, including precision, recall, and F1 score, and shows that DCGAN data augmentation improves DL model pneumonia detection

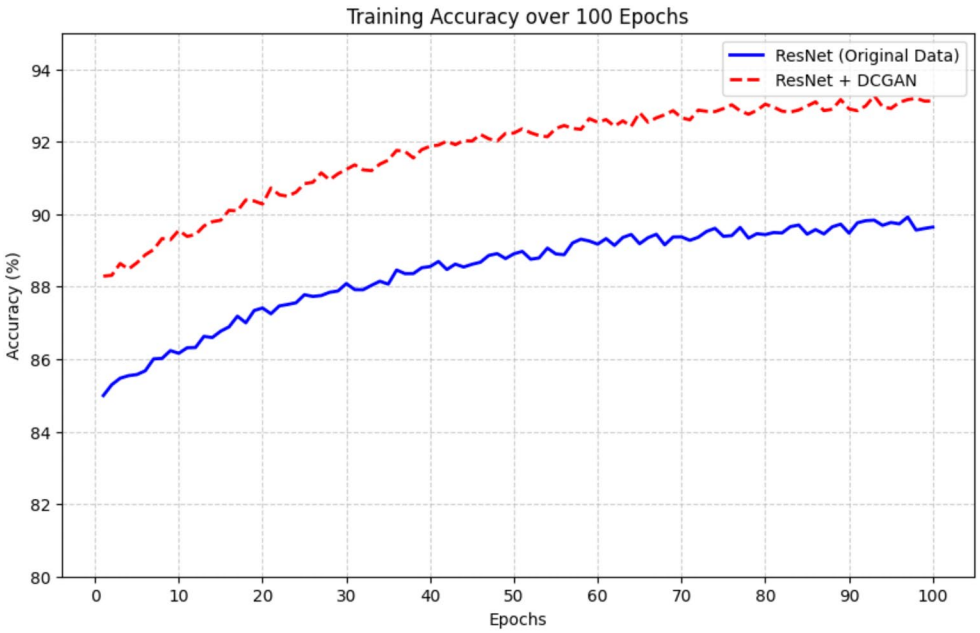


Fig. 6 Iterative Performance Graph

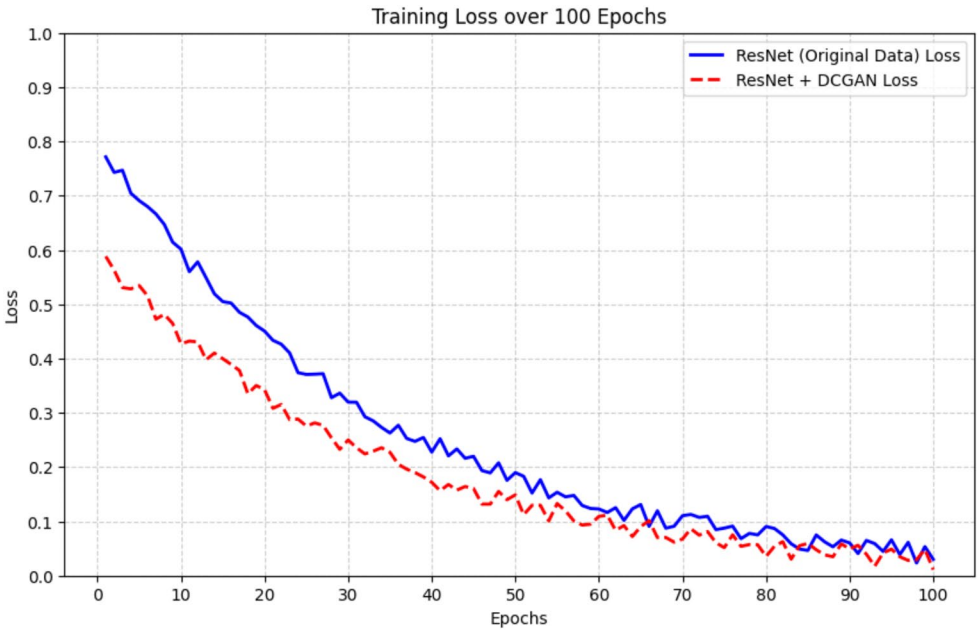


Fig. 7 Comparative loss curve

Table 2 Comparative analysis of existing state-of-the-art methods

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Remarks
Baseline CNN	82.3	79.8	81.2	80.5	Standard CNN on original data
DCGAN + CNN	85.7	83.5	84.9	84.2	Improved due to augmentation
VGG16	86.5	85.1	85.9	85.5	Stronger feature extraction
DCGAN + VGG16	89.3	88.0	88.7	88.3	Better generalisation
VGG19	87.1	85.8	86.5	86.1	Slightly better than VGG16
DCGAN + VGG19	90.0	88.7	89.3	89.0	Improved due to synthetic data
ResNet	90.1	89.5	90.0	89.7	DL performance
DCGAN + ResNet	93.4	92.8	93.2	93.0	Best performance, enhanced data

generalisation and robustness. In all these considerations, the primary importance of these findings is that DCGAN-based augmentation is useful for tackling class imbalance and enhancing medical image classification systems.

The proposed DCGAN-enhanced ResNet model outperforms traditional architectures due to its superior feature extraction capabilities and enhanced generalization. Conventional CNNs, such as Baseline CNN and VGG variants, struggle with feature representation, leading to a trade-off between precision and recall. ResNet, with its deep residual learning, mitigates vanishing gradient issues, allowing deeper network training with improved accuracy. However, DCGAN-generated synthetic samples further enhance performance by addressing class imbalance and augmenting feature diversity, which improves model robustness. Unlike standard augmentation techniques, DCGAN ensures high-quality, domain-specific sample generation, reducing overfitting while improving classification reliability. Unlike VGG16 and VGG19, which rely on deeper but non-residual architectures, ResNet's skip connections help preserve feature integrity, making learning more efficient. The 93.4% accuracy achieved by ResNet + DCGAN surpasses existing models by leveraging adversarial learning to refine feature extraction, ensuring high precision (92.8%) and recall (93.2%). This balance minimizes false positives and false negatives, making the model more reliable in real-world scenarios. Overall, the integration of DCGAN with ResNet significantly enhances feature representation, dataset diversity, and classification performance, making it superior to conventional DL architectures.

4.3 Ablation study

The study uses ablation tests to investigate how various techniques affect performance by examining different deep-learning structures combined with augmentation methods. The presented architectures display their effectiveness through the spyder (radar curve), as shown in Fig. 8. The baseline CNN model served as a basic framework because it reached an accuracy level of 82.3% when trained on the original unmodified dataset. Implementing synthetic augmentation based on DCGAN generated significant progress,

leading to an accuracy level of 85.7%. The study results indicate that data augmentation played a crucial part in balancing class distribution by providing superior performance to the minority NORMAL class. The research examined how pre-trained models performed, particularly VGG16, VGG19, and ResNet50, with their excellent feature extraction features. Due to their improved feature representations, both VGG16 and VGG19 achieved an accuracy of 86.5% and 87.1%, respectively, with VGG19 performing marginally better. ResNet50 yielded the highest result with 90.1% accuracy under the condition of no augmentation because its residual learning mechanism effectively protects deep feature representation abilities.

Researchers added generated images from DCGAN to pre-trained model training datasets for further data augmentation assessment. All models showed improving performance in the experimental outcomes. The augmentation technique resulted in a performance gain from 86.5 to 89.3% for VGG16, and VGG19 achieved 90.0% accuracy, for example. DCGAN-generated images with ResNet achieved the most significant accuracy improvement because their integration elevated ResNet50 accuracy from 90.1 to 93.4%. The use of synthetic samples during training specifically improves the generalization of complex model architectures, which need diverse data to build robustness. Experimental findings demonstrated that DCGAN-based augmentation produced better precision and recall scores and F1-score improvements among different models, demonstrating both class distribution balance and a decline in false-positive and false-negative errors. According to this ablation experiment, deep-learning medical image classification models strongly depend on selected architectures and augmentation strategies.

The 3D Precision-Recall-F1 visualization in Fig. 9 allows a comprehensive understanding of model performance by showing evaluation metric interrelationships. Data enhanced through DCGAN achieves better classification effectiveness while providing stable results that result in a distinct pattern according to the data. Different networks display distinct behavior patterns based on their precision-recall axis location. Specific architectures tend to focus either on precision reduction of false positives or recall increase to minimize false negatives. Models that obtain the best harmonic balance between precision and recall receive recognition through the F1-score evaluation. Models' position in the plot exhibits key relationships between the extraction of features and the robustness of decision boundaries. The visual depiction enables researchers to understand performance variations between different models, including deeper architecture systems, because of their superior generalization abilities. Organizational difficulties make it possible to choose the proper model that fits specific real-world needs, which optimizes classification results while maintaining reliability.

5 Conclusion and future direction

This research outlines the application of DCGANs in DL-based pneumonia detection from chest X-ray images. The augmentation analysis reveals that model performance is boosted greatly, especially with class imbalance, when DCGANs are used to produce synthetic images. Implementing augmentation through DCGANs with ResNet50 achieved the highest accuracy of 93.4%, outperforming all other tested architectures and traditional methods. The results show that DCGAN-generated images increase the variety of data and enhance generalisation, improving classification accuracy, precision, recall, and F1-score. Combining deep residual learning with synthetic data augmentation is highly effective in medical image analysis. It can be applied to other computer-aided diagnosis (CAD) systems, fostering better DL models for healthcare that are more data-efficient.

Along with the positive results, several considerations still need attention to improve the effectiveness of pneumonia detection using DCGAN augmentation. Broadening this study to include larger and more heterogeneous multi-institutional datasets may allow the generalizability of the approach. Using more sophisticated generative models such as StyleGAN, CycleGAN, or Diffusion Models can further improve the quality of synthesised medical images. Incorporating Explainable AI (XAI) approaches such as Grad-CAM or SHAP can enhance model trust, allowing the clinician to understand the DL predictions. Also, expanding the study to include multi-class pneumonia detection, where differentiation between bacterial and viral pneumonia is made, enhances the clinical benefit. Clinical validation is essential as it determines whether the model affects a radiologist's use of clinical judgment versus how much they rely on the model to make an accurate diagnosis. Combining GAN-based image augmentation with self-supervised learning or transformer models is one way to improve feature extraction and classification accuracy. Altering the model's design to decrease the need for computational resources will make it more practical in hospitals and low-resource environments, making using GANs for augmentation much more efficient. Attending to these issues will push AI-powered medical diagnostics closer to more precise, clearer, and easier-to-escalate solutions, leading to better results in detecting and treating diseases and overall health care.

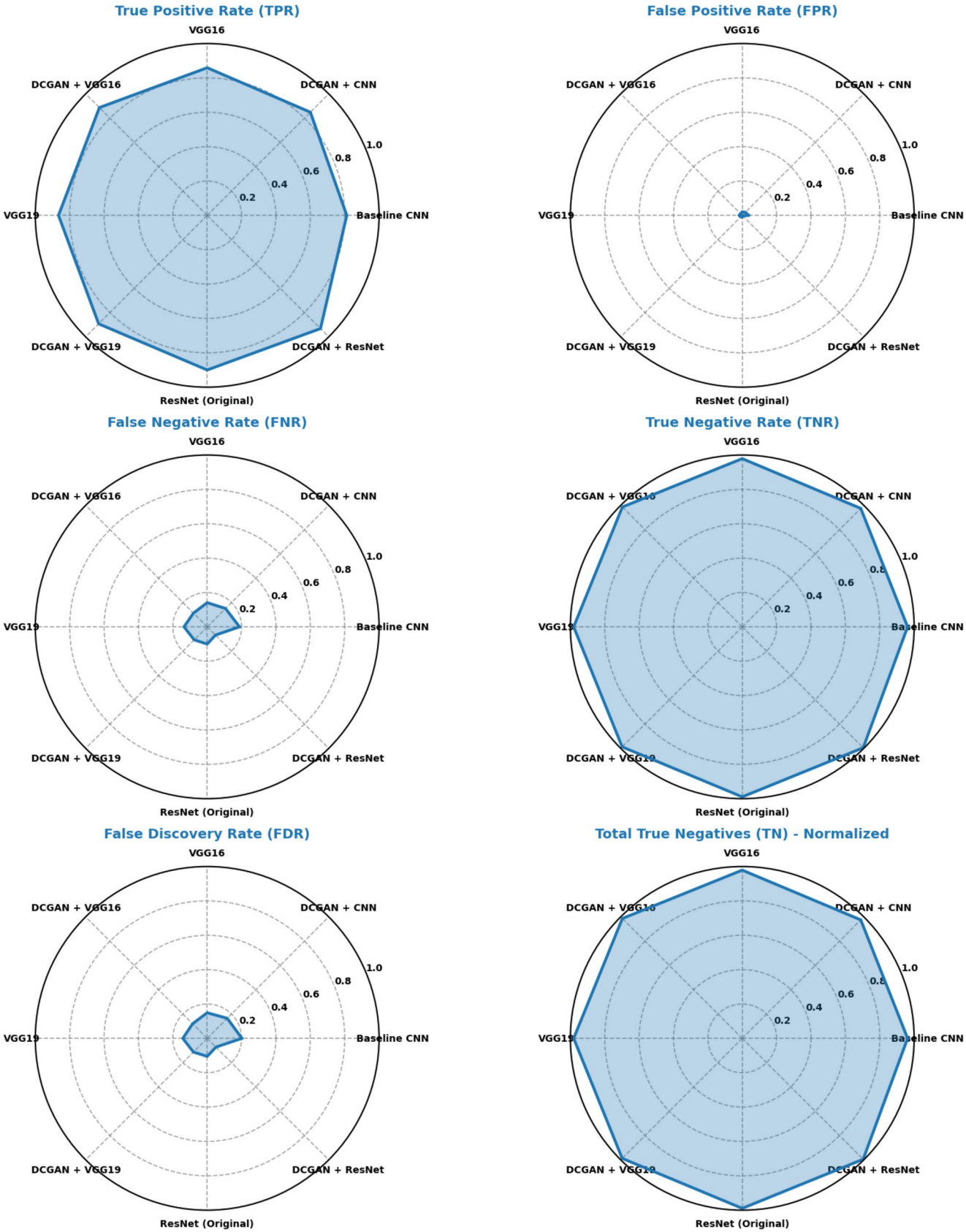


Fig. 8 Spyder curve analysis for all models

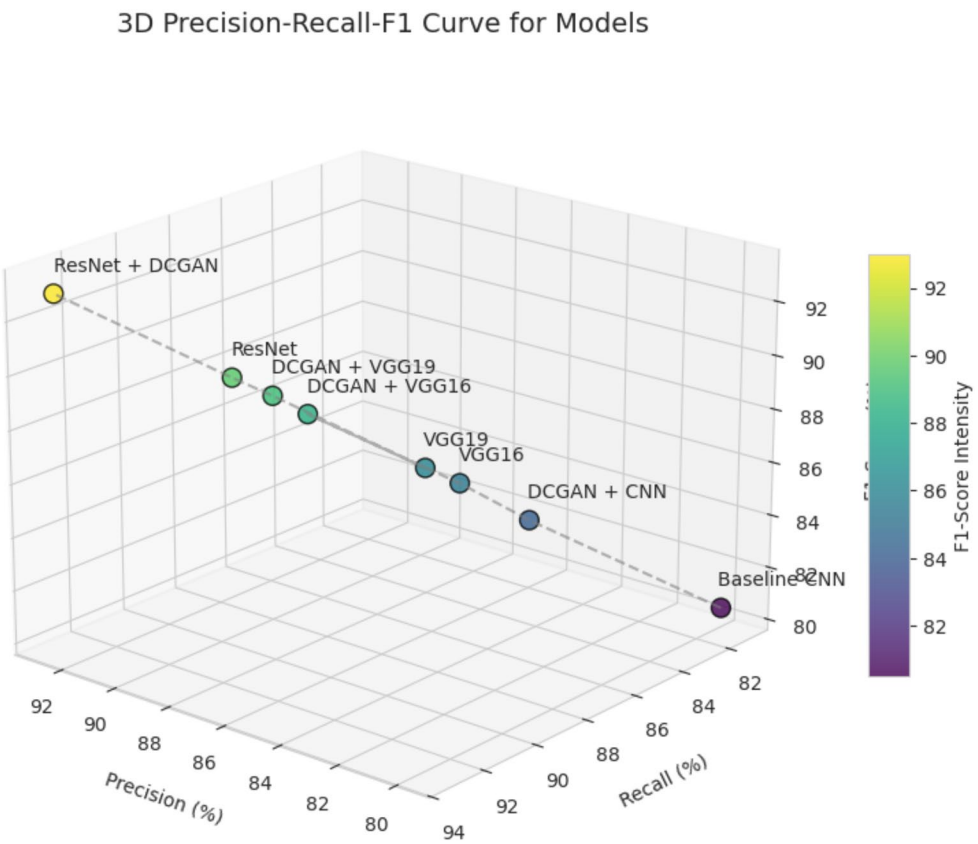


Fig. 9 3D precision-recall-f1 curve for model

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Declarations

Competing interests The authors declare that there is no conflict of interest regarding the publication of this paper.

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